Sungmoon Choi

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RESEARCH INTERESTS

Astrodynamics, Orbital Mechanics, Space Mission Design and Navigation, Trajectory Optimization, Tour Design, Multi-Body Dynamics, Global Optimization, Machine Learning, Artificial Intelligence, Contour Integral

EDUCATION

Ph.D., *Aerospace Engineering (Minor: Computer Science)*	Iowa State University, Ames, IA	2026.05
M.S., *Aerospace Engineering*	Iowa State University, Ames, IA	2025.12
M.S., *Astronomy*	Yonsei University, Seoul, Korea	2020.02
B.S., *Mechanical Engineering*	Pusan National University, Busan, Korea	2016.08

PROFESSIONAL APPOINTMENTS

New Mexico State University — Las Cruces, NM

Assistant Professor (*Tenure-Track*), Department of Mechanical and Aerospace Engineering — 2026.08 – Present

NASA Jet Propulsion Laboratory — Pasadena, CA

Research Intern, Mission Design and Navigation Section — 2023.06 – 2025.12

AWARDS & ACHIEVEMENTS

- Miller Completion Award, *Iowa State University* — 2025.12
- ISEB–KARI Scholarship awarded by the Space Generation Advisory Council *(SGAC)* — 2025.09
- Research Excellence Award, *Iowa State University* — 2025.05
- Exceptional Performance Award for GTOC Team, *NASA JPL* — 2023.09
- 1st Place at Global Trajectory Optimization Competition (GTOC) 12, *NASA JPL* — 2023.08
- Top 5 Proposal as Principal Investigator, *NASA L'SPACE Program* — 2022.08
- Outstanding Service Award, *Korean-American Scientists and Engineers Association (KSEA)* — 2022.08
- Excellent Paper Presentation Award, *Korean Society for Aeronautical and Space Sciences* — 2019.11
- IAC / ISEB / SGAC 2017 Representative of KARI, *Korea Aerospace Research Institute (KARI)* — 2017.09

SCHOLARSHIPS & FELLOWSHIPS

- The Korean Government Scholarship Program for Study Overseas
- Research / Teaching Assistantship from graduate program
- OK Savings Bank Fellowship
- Academic Excellence Scholarship
- National Scholarship
- Education capability enhancement project

JOURNAL PAPERS

2026

(J3) **Choi, S.**, Das-Stuart, A., Anderson, R. L., & Abdelkhalik, O. (2026). Vectorized dynamic-size genetic algorithm for rapid generation of optimal and suboptimal flyby sequences. *Astrodynamics*, 1-24. https://doi.org/10.1007/s42064-026-0317-5

(J2) **Choi, S.**, Negrete, A., Abdelkhalik, O., Servadio, S., & Lee, D. D. (2026). Ambiguity-Free Geometric Approach to Deep-Space Navigation Using Angles-Only Measurements. *Journal of Guidance, Control, and Dynamics*, 1-14. https://doi.org/10.2514/1.G009699

2025

(J1) **Choi, S.**, Abdelkhalik, O., & He, P. (2025). Multidisciplinary optimization of end-to-end Mars aerobraking trajectory and spacecraft design. *Advances in Space Research 75, no. 11 (2025): 8065-8083.* http://dx.doi.org/10.1016/j.asr.2025.03.022

CONFERENCE PROCEEDINGS

2026

(C12) **Choi, S**., Abdelkhalik, O., Servadio, S., & Lee, D. (2026). Deep-Space Initial Orbit Determination from Angles-Only Measurements Without Prior Attitude or Distance Information. In *AIAA/AAS Astrodynamics Specialist Conference*

2025

(C11) **Choi, S.**, Gibson, T., & Abdelkhalik, O. (2025). Learning-Based Placement of Deep Space Maneuvers in the Rendezvous Problem. In *Daniele Mortari Astronautics Symposium (URL)*

(C10) Negrete, A., **Choi, S.**, & Abdelkhalik, O. (2025). Shape-based Approximation for Low-Thrust Spacecraft Trajectory Using Chebyshev Polynomials. In *Daniele Mortari Astronautics Symposium (URL)*

(C9) **Choi, S.**, & Abdelkhalik, O. (2025). Shape-based Low-Thrust ΔV Prediction Using Deep Neural Networks. In *Daniele Mortari Astronautics Symposium (URL)*

(C8) **Choi, S.**, Das-Stuart, A., Anderson, R. L., & Abdelkhalik, O. (2025). Reinforcement Learning for Multiple Gravity Assist Trajectory Optimization. In *76th International Astronautical Congress (IAC 2025)*

(C7) **Choi, S**., Abdelkhalik, O., Das-Stuart, A., & Anderson, R. L. (2025). Neural Network-Based Universal Variable Approximation for Lambert's Problem. In *AIAA/AAS Space Flight Mechanics Meeting*

(C6) **Choi, S**., Abdelkhalik, O., & He, P. (2025). Multidisciplinary Optimization of End-to-End Mars Aerobraking Trajectory and Spacecraft Design. In *AIAA/AAS Space Flight Mechanics Meeting*

2024

(C5) **Choi, S**., Abdelkhalik, O., Das-Stuart, A., & Anderson, R. L. (2024). Vectorized Dynamic Size Genetic Algorithm for Fast Flyby Sequence Generation. In *AIAA/AAS Astrodynamics Specialist Conference*

(C4) **Choi, S**., Abdelkhalik, O., Das-Stuart, A., & Anderson, R. L. (2024). Machine Learning Application for Lambert's Problem. In *AIAA/AAS Astrodynamics Specialist Conference*

(C3) **Choi, S**., Knapick, D., Vijayakumar, M., & Abdelkhalik, O. (2024). Hidden Genes Genetic Algorithm with Low-Thrust Shape-Based Method. In *AIAA SCITECH 2024 Forum* (p. 0629)

2019

(C2) **Choi, S**., Choi, J., & Park, C. (2019). Trans-Lunar Trajectory Design Using Virtual Orbital Plane Based on Ephemeris. *Abstracts of the Korean Society for Aeronautical and Space Sciences Conference,* 214-215
Excellent Paper Presentation Award

2018

(C1) Choi, J., Hwang, J., **Choi, S.**, & Park, C. (2018). Preliminary Mission Planning for Multiple Asteroid Exploration. *Abstracts of the Korean Society for Aeronautical and Space Sciences Conference,* 313-314

SELECTED PRESENTATIONS

26.04.10	**Choi, S.** "Machine Learning-Aided Tour Design and Angles-Only Navigation for Deep-Space Missions," Seminar in the Department of Mechanical and Aerospace Engineering, New Mexico State University, *Las Cruces, NM*
26.03.05	**Choi, S.** "Machine Learning-Aided Tour Design and Trajectory Optimization," Seminar in the Department of Mechanical and Aerospace Engineering, California State University, Long Beach, *Long Beach, CA*
25.12.18	**Choi, S.** "Reinforcement Learning for Multiple Gravity Assist Trajectory Optimization," Final Presentation in Mission Design and Navigation Section at Jet Propulsion Laboratory, *Pasadena, CA*
25.08.06	**Choi, S.** "Machine Learning 101," Invited Lecture in Mission Design and Navigation Section at Jet Propulsion Laboratory, *Pasadena, CA*
24.08.05	**Choi, S.** "Neural Network-Based Lambert's Problem & Vectorized Dynamic-Size Genetic Algorithm," Final Presentation in Mission Design and Navigation Section at Jet Propulsion Laboratory, *Pasadena, CA*

POSTER PRESENTATIONS

2026.07 | (P4) Yi, A., Servadio, S., & **Choi, S.** (2026). Robust Deep-Space Navigation Via Four-Body Angles-Only Analysis. EUReCA Summer Undergraduate Research Showcase, Ames, Iowa

2025.04 | (P3) **Choi, S.**, & Abdelkhalik, O. (2025). Ambiguity-Free Geometrical Deep Space Navigation Using Angle-Only Measurements. Iowa Space Grant Consortium Spring Research Symposium, Ames, Iowa

2019.02 | (P2) **Choi, S.**, & Park, C. (2019). That's One Small Step for a Man, One Giant Leap for Mankind. The 2nd Astro Poster Jamboree. Seoul, Korea

2014.08 | (P1) **Choi, S.**, Park, H., Park, J., Chung, Y., Held, D., & Kim, D. (2014). Development of Smart Traps for Asian Ambrosia Beetles. UKC 2014*,* San Francisco, CA

STUDENT CONFERENCE PROCEEDINGS

2026

(S3) Larsen, K. (2026, March). A Contour-Integral Approach to Collinear Lagrange Points and Zero-Velocity Region Openings in the Circular Restricted Three-Body Problem. In *2026 Regional Student Conferences* (p. 114205).

(S2) Roy, T. (2026, March). Low-Thrust Trajectory Optimization via Direct Transcription Using Extreme Theory of Functional Connections. In *2026 Regional Student Conferences* (p. 114788).

(S1) Park, M. (2026, March). A Comparative Study of Physics-Informed Neural Networks and Extreme Theory of Functional Connections for Two-Point Orbital Transfers. In *2026 Regional Student Conferences* (p. 115555).

CURRENT RESEARCH PROJECTS

Reinforcement Learning for Tour Design

• Developing tour design algorithms and contingency planning with trajectory reuse, in collaboration with NASA JPL

Machine Learning Applications to Lambert's Problem

• Constructing data-driven frameworks for fast prediction of Lambert's problems, in collaboration with NASA JPL

Geometric Approach to Optical Navigation for Deep Space Missions

• Integrating unified position/attitude measurement models into state estimation, in collaboration with Iowa State University

Low-Thrust Trajectory Design Using a Shape-Based Approach

• Establishing shape-based trajectory formulation incorporating necessary conditions of optimality

Machine Learning–Based Prediction of Optimal Deep Space Maneuvers

• Applying machine learning models to predict the location and timing of optimal deep space maneuvers

Physics-Informed Neural Network Application to the Circular Restricted Three-Body Problem

• Investigating physics-based machine learning to resolve two-point boundary-value problems in CRTBP

Low-Thrust Trajectory Optimization Using Extreme Theory of Functional Connections

• Leveraging extreme theory of functional connection for efficient solution of low-thrust optimal control problems

Contour-Integral Methods in the Circular Restricted Three-Body Problem

• Computing families of periodic orbits in the CRTBP via contour-integral techniques

ADVISING AND MENTORING

Undergraduate Students

- **Tishyo Roy**, *Iowa State University* – (**S2**) 2025.09 – Present
- **Mingeun Park**, *Iowa State University* – (**S1**) 2024.12 – Present
- **Kevin Vo**, *University of Minnesota* 2026.04 – Present
- **Andrew Yi**, *University at Buffalo* – **(P4)** 2026.06 – 2026.08
- **Kimberly Larsen**, *Iowa State University* – (**S3**) 2025.09 – 2026.05
- **Ethan Griesman**, *Iowa State University* 2026.04 – 2026.06
- **Samantha Austin** (Boeing Research Fellow), *Iowa State University* 2025.10 – 2025.12
- **Timothy Gibson**, *Iowa State University* – (**C11**) 2025.02 – 2025.12
- **Nicholas Fang**, *Iowa State University* 2025.02 – 2025.05
- **Akshatra Sureshbabu**, *Iowa State University* 2024.09 – 2025.09
- **Modi Mihir** (Boeing Research Fellow), *Iowa State University* 2023.09 – 2024.06

TEACHING EXPERIENCE

Instructor, New Mexico State University

- Upcoming courses: Orbital Mechanics / System Engineering

Teaching Assistant, Iowa State University

- Mechanics of Materials Laboratory at Iowa State University *(EM 3270)* S2026, F2025
- Flight Control System at Iowa State University *(AERE 3310)* S2025, F2024, S2024, S2023

Teaching Assistant, Yonsei University

- Understanding of Space at Yonsei University *(UCC 1160)* F2019, S2019, F2018, S2018

SERVICE TO PROFESSION

Reviewer for IEEE Transactions on Aerospace and Electronic Systems

Reviewer for Astrodynamics Journal

Reviewer for IEEE Congress on Evolutionary Computation

Reviewer for AIAA SciTech